Typed Lecture Notes

Mineral Metabolism · Metalloproteins, Metalloenzymes & the Nitrogen Cycle · Hormones

Part 1 — Mineral Metabolism

Mineral metabolism covers the body's absorption, distribution, utilization and excretion of minerals (like Ca, P, Fe, Zn etc), regulated by hormones, vitamin D and diet, for functions from bone health to nerve signals, muscle function and oxygen transport.

Absorption of the minerals is primarily in the small intestine (duodenum for calcium), regulated by transporters and carrier proteins, and influenced by vitamin D. The minerals also move via blood, bound to proteins, to tissues for use.

The minerals act as cofactors (e.g. Mg), as structural components (Ca, P in bones), and are also vital for nerve impulses, muscle function etc.

Mineral Elements

The principal minerals include Ca, Mg, P, Na, K and Cl. Trace elements include Fe, Cu, Co, Zn, Mn and I. All these have shown considerable biochemical functions in the human body.

Calcium

This is the major mineral cation of bone. The body contains a very large calcium pool. A 70 kg man contains 1,200 g Ca, of which 99% is present in the bones and teeth, as calcium phosphate ($Ca_3(PO_4)_2$). The small amount of Ca not present in the skeletal structure is in the body fluids, where it is in part ionized. Much of the calcium in soft tissue is concentrated in muscle, although the membrane and cytoplasm of every cell contain some.

The roles of Ca in the ECF (extracellular fluids) and some soft tissues are so important that it will be mobilised from reserves in the bones to maintain a functional level. In addition, when intakes are low, the body can adapt by absorbing a higher proportion of dietary calcium and excreting less. These adaptations reduce the effect of intakes below recommended levels.

Functions

Some of the functions of calcium are in bone formation, blood clotting, muscle contraction, tooth formation, nerve impulse transmission, mediation of certain hormonal signals (as 2nd messenger) and as a cofactor in various enzyme reactions.

a) Bone formation

Early in foetal development a strong but flexible protein matrix or pattern of bone begins to be formed. The matrix remains rather flexible until after birth, possibly to facilitate the birth process. This matrix, which accounts for 30% of the bone, is composed of fibres of the protein collagen, embedded in a gelatinous ground substance composed of a carbohydrate (mucopolysaccharide) and a glycoprotein.

Shortly after birth this matrix gains strength and rigidity, primarily as a result of the deposition and growth of mineral crystals within the matrix, in a process known as ossification or calcification. These crystals are either calcium phosphate and/or a combination of calcium phosphate and calcium hydroxide, which makes up a physiologically stable compound called **hydroxyapatite** [$Ca_{10}(PO_4)_6(OH)_2$].

Since calcium and phosphate are the predominant mineral elements in these compounds, an adequate supply of both must be present before they can precipitate from the fluids surrounding the bone matrix to grow in the bone tissue. This deposition of calcium compounds in the mature

collagen matrix appears to begin 2 to 10 days after the collagen has been laid down by the osteocytes or living bone cells. The transfer of calcium and phosphates into the matrix is facilitated by the mitochondria of these cells.

b) Tooth formation

The mineral that forms dentine and enamel is the same hydroxyapatite found in bones, but the crystals are more dense and the water content (<10%) is lower. The protein in enamel is keratin, whereas that in dentine is collagen. In contrast to bones, which are relatively active metabolically, teeth undergo almost no change once they have erupted into the oral cavity.

c) Blood clotting

Once cells have been injured, ionized calcium, representing over half of the total blood Ca, stimulates the release of a phospholipid, thromboplastin, from the blood platelets. Thromboplastin in turn catalyzes the conversion of prothrombin to thrombin. Thrombin then aids in the polymerization of fibrinogen to fibrin, the clot. Under normal conditions blood Ca levels are maintained at a level sufficiently high to facilitate the blood-clotting process; therefore an increase in dietary calcium will have little effect on blood clotting time.

Schematic representation of the blood clotting mechanism:

Blood platelets $\rightarrow(Ca^{2+})\rightarrow$ Thromboplastin
Prothrombin $\rightarrow$(Thromboplastin, Ca^{2+})$\rightarrow$ Thrombin
Fibrinogen $\rightarrow$(Thrombin; polymerization)$\rightarrow$ **Fibrin (blood clot)**

d) Catalyst for biological reactions

The absorption of cobalamin (vitamin B_{12}) through the intestinal wall depends on Ca. Pancreatic lipase is activated by Ca, as are many of the enzymes involved in the release of energy from carbohydrates, fat and protein. The cells of the pancreas that secrete insulin must have Ca in the intracellular fluid before they can respond to the stimulation from glucose. In addition, the formation and breakdown of acetylcholine, the substance necessary for the transmission of nerve impulses from one fibre to the next, is dependent on Ca. The level of Ca needed to facilitate these reactions will be maintained at the expense of skeletal Ca; therefore, as in blood clotting, the level of dietary Ca intake will not have a direct effect on these functions.

e) Maintenance and function of cell membrane

Calcium occurs in the cell membrane closely bound to the phospholipid lecithin. Here it governs the permeability of the cell membrane to various nutrients and thus controls the uptake of nutrients by the cell.

f) Regulation of muscular contraction and transmission of nerve impulse

Factors Favouring Calcium Absorption

a) **Vitamin D:** 1,25-dihydroxycholecalciferol, formed in the kidneys, catalyzes the synthesis of a protein carrier in the mucosal cells of the intestinal wall. Calcium then attaches to the carrier, which is specific for calcium, and is transported across the cells of the intestinal membrane to the blood. Glucocorticoids secreted by the adrenal gland, and strontium, interfere with conversion of vitamin D to its biologically active form, which results in depressed Ca absorption. When vitamin D is present, Ca is absorbed throughout a greater length of the intestine than when it is absent; thus more Ca will be absorbed before the food moves to the colon, where no absorption occurs.

b) **Acidity of digestive juice:** Calcium is more soluble in acid and hence is more readily absorbed from an acid medium than from an alkaline medium. When HCl secretion is reduced, as is often the case in old age, calcium absorption may be depressed. This may be partially

compensated for by an increased intake of foods rich in ascorbic acid, which helps keep calcium ionized. The increase in calcium absorption noted on high-protein diets has been attributed to the action of some of the amino acids in forming a soluble complex with Ca to facilitate its absorption or prevent its precipitation as an insoluble complex.

c) **Lactose:** Absorption of Ca is improved in the presence of lactose. The beneficial effect of lactose in obtaining Ca from milk, a food also rich in lactose, is obvious.

Factors Depressing Calcium Absorption

a) **Oxalic acid:** Oxalic acid is an organic acid found in several fruits and vegetables. It combines in the digestive tract with calcium to form an insoluble complex, calcium oxalate, from which calcium cannot be released for absorption. In most foods containing oxalic acid, there is sufficient calcium present to tie up all the oxalic acid, leaving no surplus oxalic acid to bind Ca from other foods eaten at the same time.

b) **Phytic acid:** Phytic acid, found predominantly in the outer husks of cereals, lowers the utilization of Ca by binding it in an insoluble complex. Humans lack the enzyme phytase to hydrolyze the phytate group and to release the calcium. Only in cases in which calcium-poor foods are consumed in conjunction with foods high in phytate would this inhibitory effect become a practical problem.

Major Food Sources and Deficiency

Major food sources: Milk (cow's milk, human breast milk) and dairy products (e.g. cheese, yoghurt), leafy vegetables, legumes and nuts, and whole cereal products. Sardines and other small fish in which the bones are consumed.

Deficiency: Symptoms include tetany and related muscle and neurologic disorders.

Harmful Effects of Some Toxic Heavy Metals

1. Lead (Pb)

Lead is a toxic, non-essential heavy metal with no known biological function that causes severe health damage to all living organisms. It is a major heavy metal contaminant to food and is finding its way into our food supply in increasing amounts. Leaded fuels contribute 90% of lead in the environment. Soils contaminated by airborne lead accumulate the mineral, and plants absorb it from the soil. Animals grazing on leaded plants concentrate the mineral in their bones.

Effects on humans and animals:

1. **Nervous system:** It damages the brain and central nervous system, causing permanent cognitive impairment, behavioural and developmental disorders, lower IQ, and developmental delays in children.
2. **Blood and anaemia:** It interferes with enzymes needed to make haemoglobin and makes red blood cells fragile.
3. **Organs and systems:** It accumulates in bones for decades, damages kidney function, raises blood pressure, and causes cardiovascular problems in adults.
4. **Pregnancy:** It increases the risk of reduced fetal growth, preterm birth and miscarriages.
5. **Other:** It induces severe anaemia, digestive issues and neurological disorders in mammals and aquatic life.

2. Mercury (Hg)

Harmful effects of mercury include severe damage to the brain, nervous system, kidneys and digestive tract. The harmful effects depend largely on its chemical form, i.e. elemental, inorganic, or organic form.

i) Elemental (metallic mercury): It causes tremors, emotional changes like mood swings or nervousness, insomnia, and poor mental function and test performance. Higher exposure can lead to kidney damage, respiratory failure, and death.

Part 2 — Metalloproteins, Metalloenzymes and the Nitrogen Cycle

Metalloproteins

These are proteins containing metal ions. The metal ions (as cofactors) are essential for the structural, biological and catalytic functions of the proteins. These proteins are important for many biological processes, including oxygen transport, enzyme catalysis and cellular defense. Examples of these proteins include haemoglobin containing iron, and carbonic anhydrase containing zinc. Many of the metalloproteins are enzymes that use metal ions for their catalytic functions.

These proteins are involved in a variety of functions like (1) oxygen transport and (2) electron transfer.

1. Oxygen transport

For oxygen transport, haemoglobin is involved. It is an iron-containing protein that transports oxygen in the red blood cells of all vertebrates, while myoglobin transports oxygen in the muscle.

Haemoglobin is a conjugated protein like all other metalloproteins. It is made up of four subunits. It has a quaternary structure, which refers to the way the polypeptide chains are packed together in proteins that contain more than one polypeptide chain. The individual chains are called protomers, monomers or subunits.

It is a tetrameric protein containing four protomers, monomers or subunits, made up of 2 α chains of 141 amino acids and 2 β chains of 146 amino acids. Each subunit holds a ferrous ion (Fe^{2+}) that binds one O_2. The ferrous ion (Fe^{2+}) in haemoglobin facilitates oxygen binding and release.

So a haemoglobin molecule can carry four molecules of oxygen, essential for energy production and overall cell function.

Sickle cell haemoglobin: In sickle cell haemoglobin, the β-chain has glutamic acid at position 6 replaced with valine. This replacement results in a change of electrical charge on the haemoglobin molecule. Thus sickle cell haemoglobin is less soluble in the deoxygenated state and precipitates out, giving the red blood cells a saucer or sickle shape.

2. Electron transfer

For electron transfer, metalloproteins such as cytochromes are involved. Example: cytochrome oxidase, which is a copper- and iron-containing enzyme involved in cellular respiration.

Metalloenzymes

Metalloenzymes are enzymes that contain tightly bound metal ions, such as zinc, iron or copper, which are crucial for their catalytic activity. The metal ions are tightly bound to their active sites. They perform a variety of functions, including electron transfer, oxygen atom transfer and substrate activation, by using the metal ions to facilitate reactions.

Examples of metalloenzymes:

i) **Superoxide dismutase:** an enzyme that protects cells from oxidative damage by neutralizing reactive oxygen species. It uses manganese ion as metal cofactor.

ii) **Carbonic anhydrase:** a zinc-containing enzyme that catalyzes the hydration of carbon dioxide to bicarbonate.

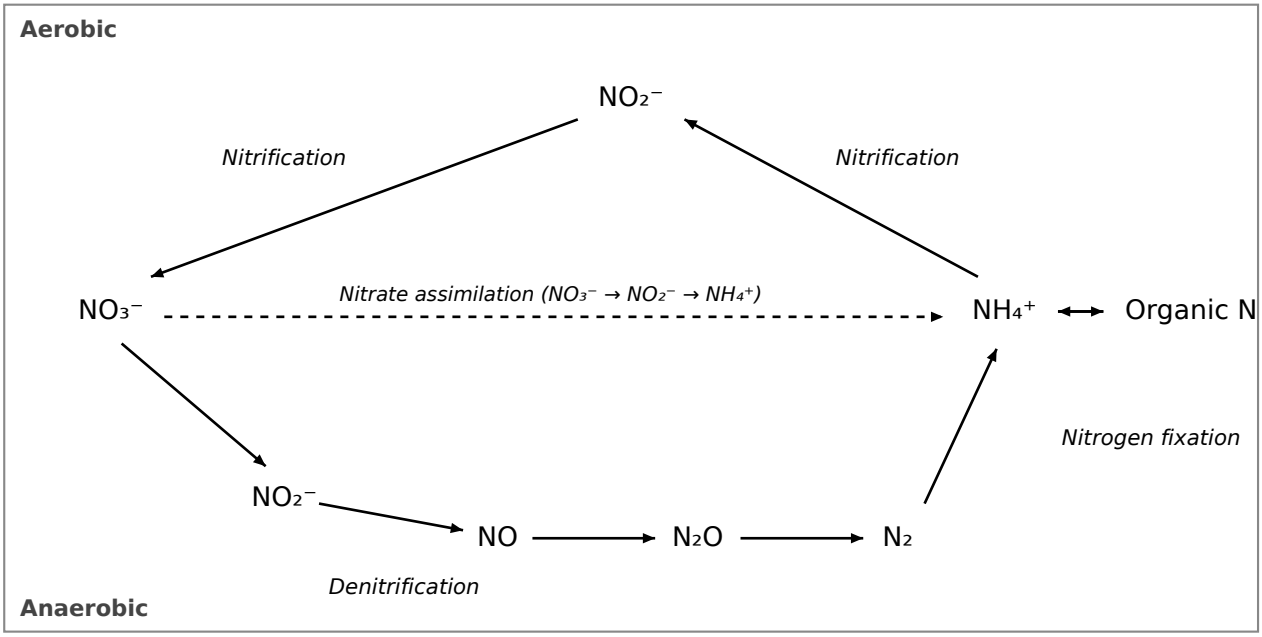
iii) **Nitrogenase:** contains a cluster of iron atoms around a molybdenum ion that is essential for converting atmospheric nitrogen into ammonia.

iv) **Cytochrome oxidase:** a copper- and iron-containing enzyme that is a key component of the electron transport chain.

v) **Carboxypeptidase:** a zinc metalloenzyme that plays a role in protein digestion.

Nitrogen Cycle

Nitrogen is cycled between organisms and the nonliving environment. Acquisition of nitrogen by biological systems is accompanied by its reduction to ammonium ion (NH_4^+) and the incorporation of NH_4^+ into organic linkage as amino or amido groups.



The nitrogen cycle

The reduction of NO_3^- to NH_4^+ occurs in green plants, various fungi and certain bacteria in a two-step metabolic pathway known as **nitrate assimilation**. The formation of NH_4^+ from N_2 gas is termed **nitrogen fixation**.

Nitrogen fixation is an exclusively prokaryotic process, although bacteria in symbiotic association with certain green plants also carry out nitrogen fixation. No animals are capable of either nitrogen fixation or nitrate assimilation, so they are totally dependent on plants and microorganisms for the synthesis of organic nitrogenous compounds, such as amino acids and proteins, to satisfy their requirements for this essential element.

Animals release excess nitrogen in a reduced form, either as NH_4^+ or as organic nitrogenous compounds such as urea. The release of nitrogen occurs both during life and as a consequence of microbial decomposition following death. Various bacteria return the reduced forms of nitrogen back to the environment by oxidizing them. The oxidation of NH_4^+ to NO_3^- by nitrifying bacteria provides the sole source of chemical energy for the life of these microbes.

Nitrate nitrogen also returns to the atmosphere as N_2 as a result of the metabolic activity of **denitrifying bacteria**. These bacteria are capable of using NO_3^- and similar oxidized inorganic forms of nitrogen as electron acceptors in place of O_2 in energy-producing pathways. The NO_3^- is reduced ultimately to dinitrogen (N_2). These bacteria thus deplete the levels of combined nitrogen, that is, nitrogen joined with other elements in chemical compounds.

Nitrate assimilation as the principal pathway for ammonium biosynthesis

Nitrate assimilation occurs in two steps:

i) the two-electron reduction of nitrate to nitrite, catalyzed by **nitrate reductase**, and

ii) the six-electron reduction of nitrite to ammonium, catalyzed by **nitrite reductase**.

$$\text{i) } NO_3^- + 2H^+ + 2e^- \rightarrow NO_2^- + H_2O$$
$$\text{ii) } NO_2^- + 8H^+ + 6e^- \rightarrow NH_4^+ + 2H_2O$$

Nitrate assimilation is the predominant means by which green plants, algae, and many microorganisms acquire nitrogen. The pathway of nitrate assimilation accounts for more than 99% of the inorganic nitrogen (nitrate or N_2) assimilated into organisms.

Nitrogen fixation

Nitrogen fixation involves the reduction of nitrogen to NH_4^+ via an enzyme system found only in prokaryotes. The heart of the nitrogen fixation process is the enzyme known as **nitrogenase**, which catalyzes the reaction:

$$N_2 + 10H^+ + 8e^- \rightarrow 2NH_4^+ + H_2$$

Reduction of two protons to hydrogen gas accompanies the biological reduction of nitrogen to ammonia. Less than 1% of the inorganic nitrogen incorporated into organic forms by organisms can be attributed to nitrogen fixation; however, this process is the only way that organisms can tap into the enormous reservoir of N_2 in the atmosphere.

Although nitrogen fixation is exclusively prokaryotic, N_2-fixing bacteria may be either free living or living as symbionts with higher plants. For example, *Rhizobia* are bacteria that fix nitrogen in symbiotic association with soybeans and other leguminous plants. This can be an important factor in plant growth and agriculture.

Nitrogen-fixing bacteria have four requirements:

1. the enzyme nitrogenase,
2. a strong reductant, such as reduced ferredoxin,
3. ATP, and
4. O_2-free conditions.

Read about: (1) the role of the nitrogenase enzyme in nitrogen fixation; (2) the sulfur cycle.

Part 3 — BCH 4113 Bioinorganic Chemistry: Hormones

Introduction

A compound is considered a hormone if:

i) it is produced at a site different from its site of action;

ii) it is carried through blood to the site of action;

iii) it controls and catalyses various metabolic processes in cells different from the cell which secretes it.

There are 3 classes of hormones: (i) peptide hormones, (ii) amine hormones and (iii) steroid hormones.

i) **Peptide hormones** are those which have from 3 to over 200 amino acid residues. They include all the hormones of the hypothalamus and pituitary as well as insulin and glucagon of the pancreas.

ii) **Amine hormones** are small, water-soluble compounds containing NH_2 groups. Examples are adrenaline of the adrenal medulla and the thyroid hormones.

iii) **Steroid hormones** are fat-soluble compounds. Examples are adrenal cortical hormones, androgens (male sex hormones) and oestrogens (female sex hormones).

The activity of hormones is regulated by 5 factors

1. The rate of synthesis at the endocrine glands and the rate of secretion from the storage granules of the endocrine.
2. In some, the specific transport system in the plasma.
3. The rate of conversion of the hormone to the more active form.
4. The hormone receptors in the target cell.
5. The rate of degradation of the hormone.

Mechanisms of Action of Hormones

1. **Induction of enzyme synthesis at the nuclear level** by stimulating transcription (mRNA). Hormones do not act directly on the DNA. Receptor proteins on the cell membrane (or in the cell cytosol) couple with the hormone, and the complex acts on the nucleus to stimulate transcription. The mRNA causes enzyme synthesis, and the enzyme causes changes in metabolism.
2. **Stimulation of translation** at ribosomal level.
3. **Direct activation at the enzyme level**, either activating or inhibiting the enzyme already present.
4. **Hormonal action at the membrane level**, e.g. insulin increases the membrane permeability to glucose.
5. **Hormonal action related to cAMP.** cAMP is synthesized from ATP by the action of adenylate cyclase, a plasma-membrane-bound enzyme, and inactivated by a second enzyme, phosphodiesterase.

$$\text{ATP} \rightarrow (\text{adenylate cyclase; } PP_i \text{ released}) \rightarrow \text{cAMP} \rightarrow (\text{phosphodiesterase}) \rightarrow 5'\text{-AMP}$$

The hormone–receptor complex either inhibits or activates adenylyl cyclase, changing the levels of cAMP. There is tissue specificity for different hormones, e.g. glucagon stimulates adenylyl cyclase activity in the liver and not in adipose tissue, while epinephrine stimulates adenylyl

cyclase in adipose tissue more than in the liver.

The cAMP produced then activates a number of enzymes known as protein kinases (phosphokinases), which in turn catalyze the phosphorylation of a number of other enzymes involved in metabolism. The phosphorylation may either activate or inactivate an enzyme. Thus cAMP acts as a **2nd messenger**, an intracellular modulator of the intracellular effector, the hormone or the **1st messenger**. The phosphorylation of phosphorylase and glycogen synthetase in the liver under the influence of glucagon or epinephrine is the best-studied example of such a mechanism.

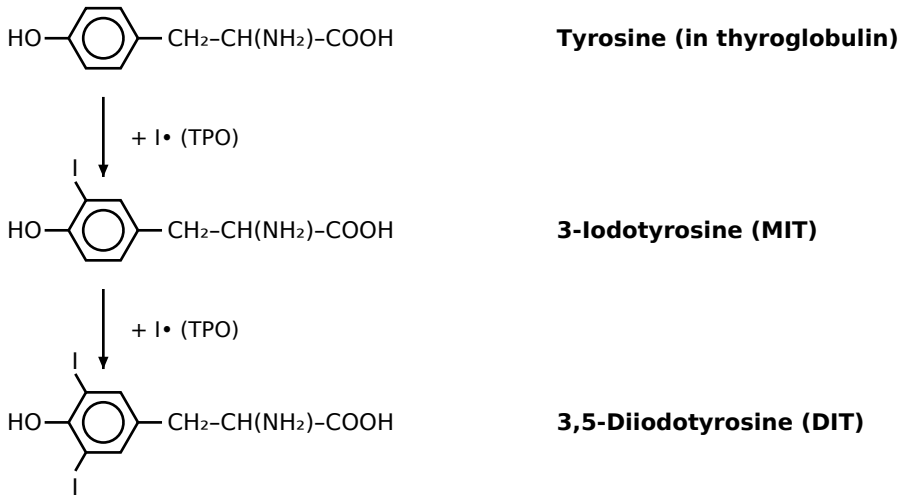
Thyroid Hormones (Iodothyronines)

The thyroid hormone thyroxine (tetraiodothyronine, T_4) and its active form triiodothyronine (T_3) are derived from the amino acid tyrosine. The following steps are involved in their biosynthesis.

1. **Iodine uptake.** The source of iodine is dietary iodine, which is Na^+ or K^+ iodide. It is absorbed from the intestines and concentrated in the thyroid gland. The thyroid uptake of iodine is stimulated by the thyroid stimulating hormone (TSH).
2. **Oxidation.** This is initiated by thyroid peroxidase (TPO), which uses H_2O_2 to oxidize iodide to iodine radical.

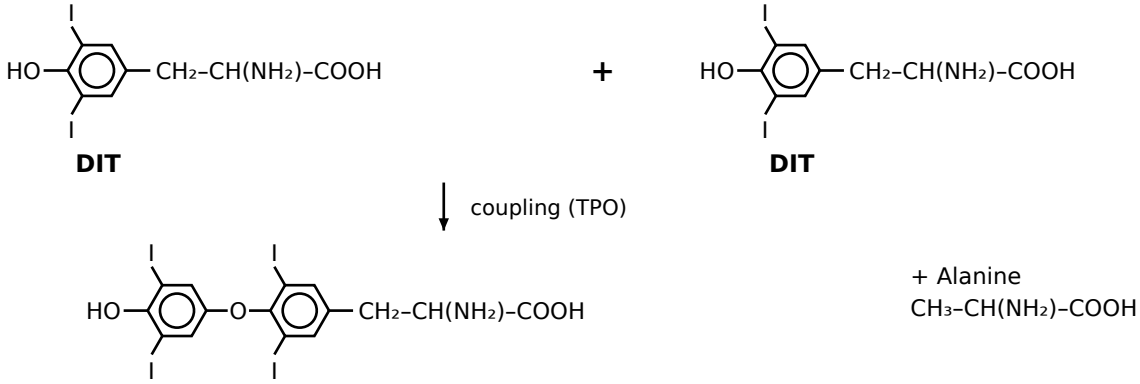
$$I^- + H_2O_2 + 2H^+ \rightarrow 2H_2O + I\bullet \text{ (iodine radical)}$$

3. **Organification.** TPO incorporates the iodine radicals onto tyrosine residues within the thyroglobulin molecules. This is known as organification.



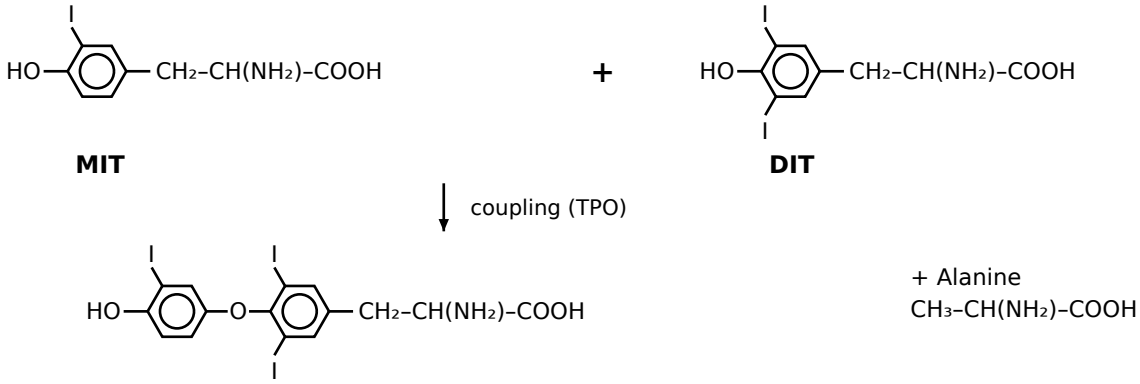
4. **Coupling.** This is also carried out by TPO: (a) DIT + DIT → T_4 (thyroxine); (b) DIT + MIT → T_3 (triiodothyronine).

(a) Coupling of two DIT molecules



3,5,3′,5′-Tetraiodothyronine (T_4, thyroxine)

(b) Coupling of MIT and DIT



3,5,3′-Triiodothyronine (T_3)

5. **Release.** A protease reaction hydrolyses the thyroglobulin, and the hormones T_3 and T_4 are released into the blood. Residual MIT and DIT are deiodinated by iodotyrosine dehalogenase, and the iodine is reutilized for further organification.

T_3 has more hormonal activity than T_4 (usually 3–5 times more active). But in the plasma, T_3 constitutes only ¼ of T_4. For the tissues, T_3 is functionally the more important hormone because it is more available to them.

Metabolic functions of thyroid hormones

1. The primary function of thyroid hormones is to regulate thermogenesis and maintain basal metabolic rate (BMR).
2. Excess thyroid hormone causes proteolysis and negative nitrogen balance.
3. They are important catalysts for protein synthesis.
4. They increase intestinal absorption of glucose.

Abnormalities of thyroid function

There are 2 types:

1. **Hypothyroidism:** insufficient amounts of T_3 and T_4 due to inborn errors of biosynthesis or thyroid failure. It could also be due to diseases of the pituitary or hypothalamus.
2. **Hyperthyroidism (thyrotoxicosis):** due to excessive production of thyroid hormones, which could be due to a tumour of the thyroid gland.

1. Hypothyroidism is of 4 types

(i) Cretinism, (ii) juvenile myxoedema, (iii) myxoedema, (iv) endemic goitre.

i) **Cretinism:** Under this condition, the thyroid gland is not developed or is totally absent. The result is that the child will be hypothyroid from birth, and the condition is known as cretinism. The child suffers from mental retardation.

ii) **Juvenile myxoedema:** This is hypothyroid activity in children. The thyroid hormone will be active in the early age when the body is small, and as the body size increases the hormonal activity does not increase (hormonal activity is not proportional to body size); as such the child will start to suffer from hypothyroid activity.

iii) **Myxoedema:** This is hypothyroidism in adults. The hypothyroid activity will continue up to the adult stage when it will no longer cope with the body requirement of thyroxine, so the adult person's mental and physical vigour will be low.

iv) **Endemic goitre:** This is the enlargement of the thyroid gland due to iodine deficiency. The decreased production of the thyroid hormone causes overstimulation or hyperactivity of the pituitary gland to secrete more thyroid stimulating hormone (TSH) and consequent enlargement of the thyroid gland.

Low I_2 in diet → low thyroid hormone in blood → high TSH from pituitary → stimulates and enlarges thyroid gland

Endemic goitre can be treated by increasing the supply of I_2 in the diet. Table salt is the usual dietary source of I_2. All governments in the world have made it compulsory that table salt should be mixed with NaI to a particular level.

2. Hyperthyroidism

There are 2 major types of hyperthyroid conditions: (i) toxic goitre and (ii) exophthalmic goitre.

i) **Toxic goitre:** During this condition, enlargement of the thyroid gland is accompanied by the secretion of excessive amounts of thyroid hormone, i.e. hyperthyroidism together with enlargement (goitre).

ii) **Exophthalmic goitre:** There is also enlargement of the thyroid gland under this condition. Symptoms include nervousness, fatigue, loss of weight, increased body temperature with excessive sweating, and an increase in heart rate. A characteristic protrusion of the eyeballs (exophthalmos) usually accompanies this hyperthyroid condition.

Hyperthyroidism can be treated by thyroid removal during surgery. It is also treated by antithyroid drugs (goitrogens), which prevent the gland from incorporating inorganic iodide into organic form (organification).

Parathyroid Hormone (PTH)

This is a single polypeptide consisting of 84 amino acids, with alanine at the N-terminus and glutamic acid at the C-terminus. The amino acid composition of the hormone between different species differs only slightly.

The hormonal activity remains with the polypeptide fraction of 34 amino acids at the N-terminus; the remaining 50 are without hormonal activity. During biosynthesis in the parathyroid gland, the hormone is synthesized as a **preprohormone** of 115 amino acids. Methionine is the N-terminal amino acid. Within five minutes after biosynthesis, 25 amino acids at the N-terminus are removed, leaving a **prohormone** of 90 amino acids. This is converted to the parathyroid hormone (PTH) immediately before secretion by the removal of 6 amino acids.

Preprohormone (115 aa) →(−25 aa)→ Prohormone (90 aa) →(−6 aa)→ PTH (84 aa)

Functions of PTH

Maintenance of plasma Ca^{2+} concentration is the most important function of PTH. This is mainly achieved by the removal of Ca^{2+} from the bone, increasing its intestinal absorption and also reducing its urinary excretion.

Calcium ions are removed from the bone by the action of vitamin D (cholecalciferol), which is hydroxylated in the liver at position 25 without involving PTH. In the kidney, 25-hydroxycholecalciferol is again hydroxylated into 1,25-dihydroxycholecalciferol (DHCC), which is the metabolically active form of vitamin D. Production of 1,25-dihydroxycholecalciferol (DHCC) is controlled by PTH in the kidneys. DHCC increases the Ca^{2+} absorption from the bone, increases the intestinal absorption of Ca^{2+} and also reduces its urinary excretion.